EXPERIMENT – 11

Performing IP Spoofing Using a GitHub Tool in Kali Linux (Educational Demonstration)

Aim

To demonstrate IP spoofing techniques using an open-source GitHub tool in Kali Linux for understanding network-level attack concepts and security vulnerabilities.

⚠️ Important Note

This experiment is conducted **strictly for academic and ethical purposes** on authorized networks only.
Unauthorized IP spoofing is illegal and punishable by law.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Tool Source:** GitHub (IP Spoofing Tool – Educational Use)
- **Programming Language:** Python / Bash (as per tool)
- **System Requirements:**
  - RAM: Minimum 4 GB (8 GB recommended)
  - Disk Space: Minimum 30 GB
  - Processor: Intel/AMD with Virtualization support

Introduction

IP spoofing is a technique where an attacker forges the source IP address in network packets to impersonate another system. It is commonly used in attacks such as DDoS, session hijacking, and bypassing IP-based authentication. This experiment demonstrates IP spoofing using a publicly available GitHub tool to understand how attackers manipulate packet headers. Kali Linux provides a safe and controlled environment for ethical network security experimentation.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

Step 2: Open Terminal

1. Launch Terminal in Kali Linux.
2. Update system packages:
3. sudo apt update

Step 3: Clone IP Spoofing Tool from GitHub

1. Clone the repository:
2. git clone https://github.com/<username>/<ip-spoofing-tool>
3. Navigate to the tool directory:
4. cd ip-spoofing-tool

Step 4: Install Dependencies

1. Install required dependencies:
2. sudo apt install python3
3. pip3 install -r requirements.txt

Step 5: Configure the Tool

1. Open the script file:
2. nano spoof.py
3. Configure:
    - Target IP address
    - Spoofed source IP address
    - Network interface

Step 6: Execute IP Spoofing

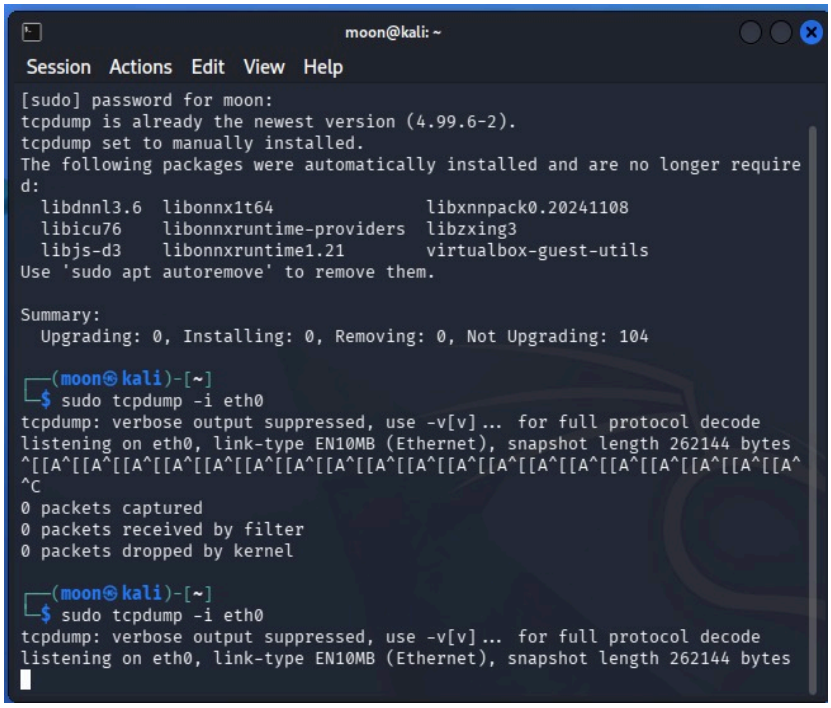
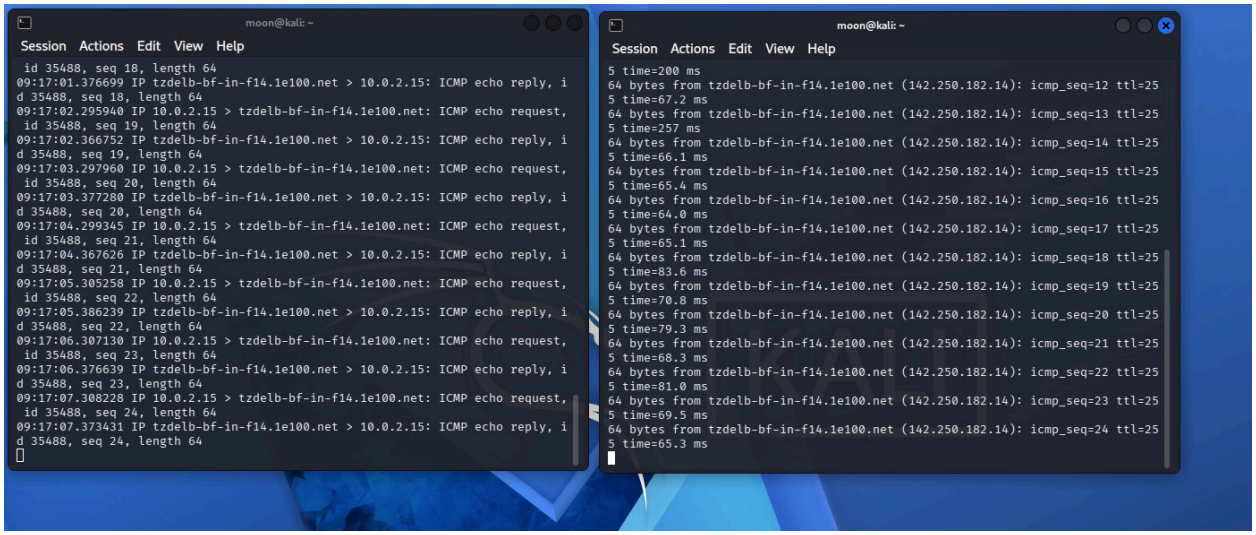
1. Run the tool with root privileges:
2. sudo python3 spoof.py
3. Observe packet transmission with spoofed IP addresses.

Step 7: Monitor Network Traffic (Optional)

1. Use Wireshark or tcpdump:
2. sudo tcpdump
3. Verify spoofed source IP packets.

Output

- Packets are sent with forged source IP addresses.
- Network traffic reflects spoofed IP information.
- Demonstration successfully completed.

Result

IP spoofing was successfully demonstrated using a GitHub-based tool in Kali Linux.
The experiment illustrates how attackers can manipulate IP headers and highlights the need for robust network security mechanisms.